



QC600 / QC600S

Motor output: 0.75 kW

Motor output: 2.2 kW

Multifunction control
unit with INVERTER



TECHNICAL DETAILS

Installation	Vertical on a vibration free and flat wall
Temperature range (operating)	-10...+50°C
Humidity	Up to 93% RH non-condensing.
Vibration	Low-vibration installation, wall mounted.
Dimension	445 x 215 x 150 mm
Supply voltage:	1~ 230VAC; 50/60 Hz; +/- 10% L1, N, GROUND Mains fuse max: 3 x 10A Rated insulation voltage U_i = 400V
Transformer	Max 13 VA , VDE 0570/EN61558 Primary 230VAC winding is thermal protected by built-in thermal transformer fuse. Both secondary windings are overload protected by multifuses.
Motor output	Max motor load by 3~ 230VAC: 2.2 kW (CB600S) / 0.75 kW (CB600) Max motor current: 10 A
Emergency stop, Stop, Thermo spec. door stop and Safety chain	Function as normal stop command and disconnect power to contactor coils
24VDC Output (terminals X3-18,X3-19)	24VDC $\pm$ 20% (non-regulated), Max load: 250mA
Safety edge input	PNE/air switch Electric type - 8k Ω termination $\pm$ 10% Optical type Performance level C, Category 2
Optical safety edge	Input voltage high (green): 2.5 - 5.0 Volt. Input voltage low (green): < 0.5 Volt. Input frequency range (green): 250 - 2000 Hz (50% duty-cycle) Pulse interval maximum (green): 7.0 mS (when not 50% dutycycle)
Photo input	X1 Terminals 9,10,11,12, 13 for 2 couple external photo, 24VDC (e.g. self contain photo cell) Performance level C, Category 2
Electronic limits	RS485, Data+ Data-, terminated with 120 Ohm
Relé (AUX1)	Max 230 Vac / 5 A (X6)
Relé (AUX2)	Electrobrake relé - NO / NC - 230V~/5A (X6)
Relé (AUX3)	Solid state relé - NO - max 30V/50mA

INVERTER FEATURES

Inverter Motor setup features			
Rated motor power 0	.1 – 2.2 kW (3 phase motor)	Rated rotary speed	500 – 4000 RPM
Rated motor voltage	100 – 400 VAC R	ated motor frequency	50 – 200 Hz
Rated motor current 1	– 10 A		
Inverter OPEN direction		Inverter CLOSE direction	
Freq. High OPEN	10 – 200 Hz	Freq. High CLOSE	10 – 200 Hz
Freq. Low OPEN 5	– 200 Hz F	req. Low CLOSE 5	– 200 Hz
Acc. Time OPEN 0	.1 – 10 Sec. A	cc. Time CLOSE 0	.1 – 10 Sec.
Dec. Time full to Low 0	.1 – 10 Sec. D	ec. Time full to Low 0	.1 – 10 Sec.
Dec. Time to stop	0.1 – 10 sec. D	ec. Time to stop 0	.1 – 10 sec.
Low speed setpoint before OPEN limit	5 – 50 % of run range L	ow speed setpoint before CLOSE limit	5 – 50 % of run range

DIRECTIVE

2014/35/UE 2014/30/CE 2014/53/CE 2006/42/CE

EN 61000-6-2:2005 EN 61000-6-3:2007 + A1:2011

EN 60950-1:2006 + A1:2010 + A2:2013 +A11:2009 + A12:2011

EN 13241-1:2004 + A1:2011 EN 13849-1:2015

EN 12453:2017 EN 12978:2003 + A1:2009

EN 301489-1:2008 EN 301489-3:2002

EN 300220-2:2007 EN 13241-1:2004 + A1:2011

IMPORTANT SAFETY INSTRUCTIONS

These instructions contain important information on the installation and the use of the CB600/CB600S unit. Please keep them even after installation. Do not start installing the CB600/CB600S unit without having first read these instructions. The installation can only be made by a qualified technician. In order to conform to the rules in force, the installation must include an automatic cut off switch with minimum distance between the contacts of at least 3mm. The safety of the control unit depends on a correct installation. Be at a far distance from shutter/awnings in movement. Check frequently if any loss of balance, sign of wear or damages wires are shown. Do not use if any repairing or maintenance is needed.

Gaposa declines all responsibility in case of improper use of the product or of an incorrect installation.

WARNING: it is very important for the people's safety to follow strictly these instructions.

IMPORTANT: Keep beholding the shutter and the people at distance while this is in movement.

IMPORTANT: Frequently check whether any cable unbalancing, wear out or damage happens. Do not use the control unit in case a reparation or a maintenance job is required.

KEEP THIS INSTRUCTIONS.

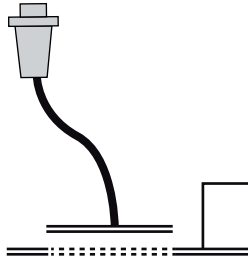
Warning: it is highly recommended to activate the impulsive mode only after having completed the set-up and adjustments of the control unit. In particular, during the mechanical limit switches adjustment select always the dead-man operation mode. During the encoder limit switches set-up the control unit will only allow the dead-man working mode.



NEVER TOUCH THE INTERNAL ELEMENTS WITHIN 15 MINUTES AFTER POWER OFF. WAIT TILL IT IS COMPLETELY DISCHARGED.

MAIN POWER 1 PHASE 230VAC

PE L1 N

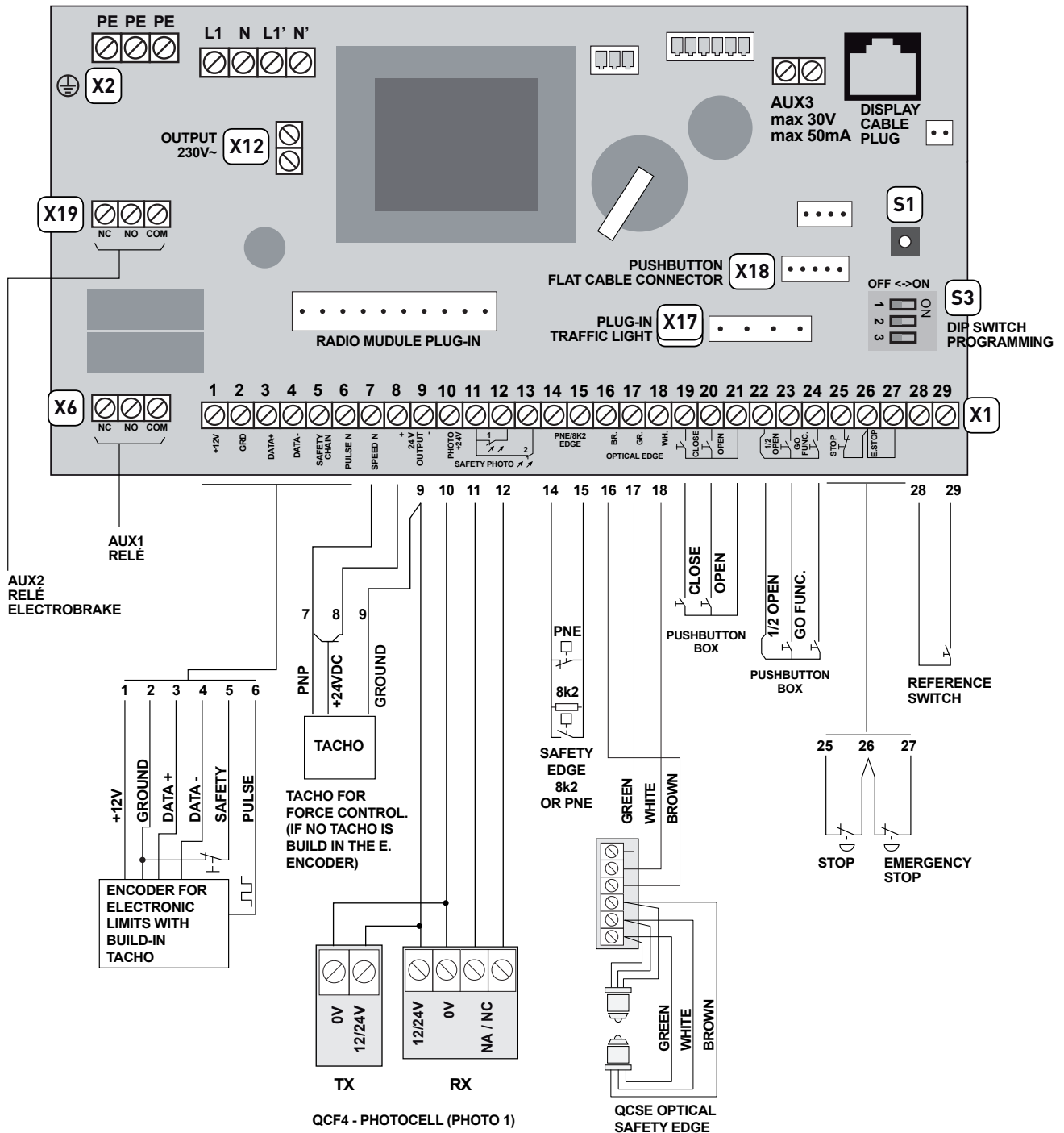
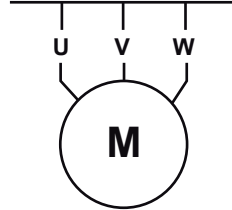


POWER TO INVERTER

PE L1' N'

INVERTER

U V W



1 ENCLOSURE INSTALLATION

For a correct installation:

- Install where the control unit can be protected from rain or adverse weather conditions
- Mounting must be vertical
- The surface has to be checked for flatness, slope and freedom from vibrations
- Do not install in an area of potential risk of condensation
- It is important that the door can be clearly seen from the position of the control through its travel
- Install in an area not accessible to children or unauthorized persons
- Do not perform any electrical connections before the enclosure installation is completely accomplished

2 ELECTRICAL OPERATING INSTRUCTIONS

(Read carefully and respect the connection's sequence)

IMPORTANT! All the connection operations must be performed only after the main power supply has been disconnected.
TURN OFF THE MAIN POWER SWITCH BEFORE ANY OTHER OPERATION!



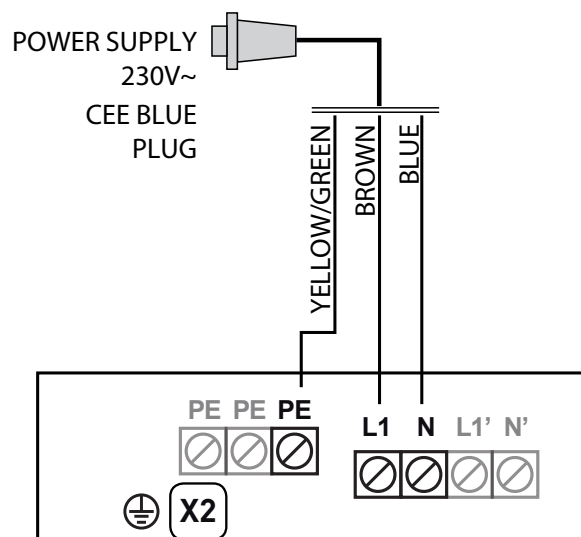
**NEVER TOUCH THE INTERNAL ELEMENTS WITHIN 15 MINUTES AFTER POWER OFF.
WAIT TILL IT IS COMPLETELY DISCHARGED.**

2.1 CONTROL UNIT POWER SUPPLY

WARNING! The installation must include an automatic cut off switch with minimum distance between the contacts of at least 3 mm.

WARNING: if you power a motor with a power mode other than that for which it is set you can damage the motor and the control unit and put at risk the safety of the installer.

Here below you will find the power supply diagrams:



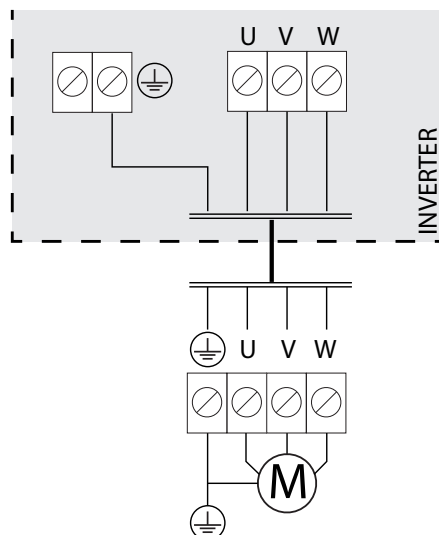
If you need to disconnect the power cable and then to reconnect it or change the control unit wiring sequence, you **HAVE TO** connect the wires properly, restoring the original configuration.

WARNING: if you connect the wires differently from what is shown in the diagrams you can damage the motor and the control unit and endanger the safety of the installer.

2.2 CONNECTING THE CONTROL TO THE MOTOR

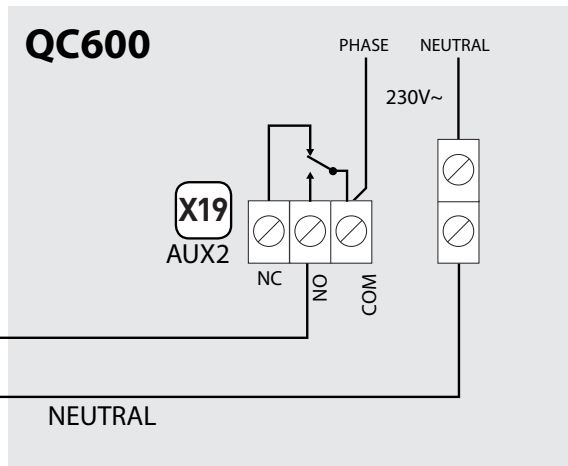
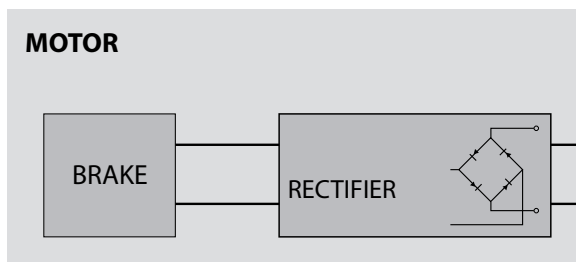
IMPORTANT! All the connection operations must be performed only after the main power supply has been disconnected. **TURN OFF THE MAIN POWER SWITCH BEFORE ANY OPERATION!**

Here below you will find the power supply diagrams



2.3 ELECTROBRAKE CONNECTION

Connect one of the terminals of the electric brake wire to neutral connection straightener (flying clamp) and the other Terminal to the Terminal AUX2 the terminal box X6:



3 INTEGRATED PUSHBUTTON

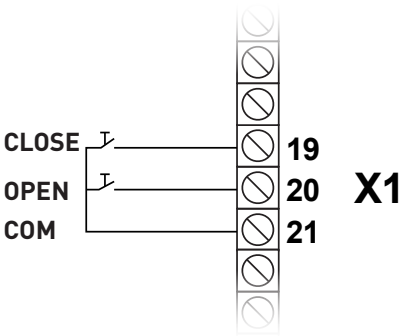
The integrated pushbutton on the cover of the control unit is connected to "DISPLAY RS485" connector



3.1 ADDITIONAL CONTROL PUSHBUTTONS

You can connect additional control pushbuttons through the terminals from 19 to 21 of X1 connector.

- connect a normally open button to the contacts [21] and [20] for the OPEN command;
- connect a normally open button to the contacts [21] and [19] for the CLOSE command;



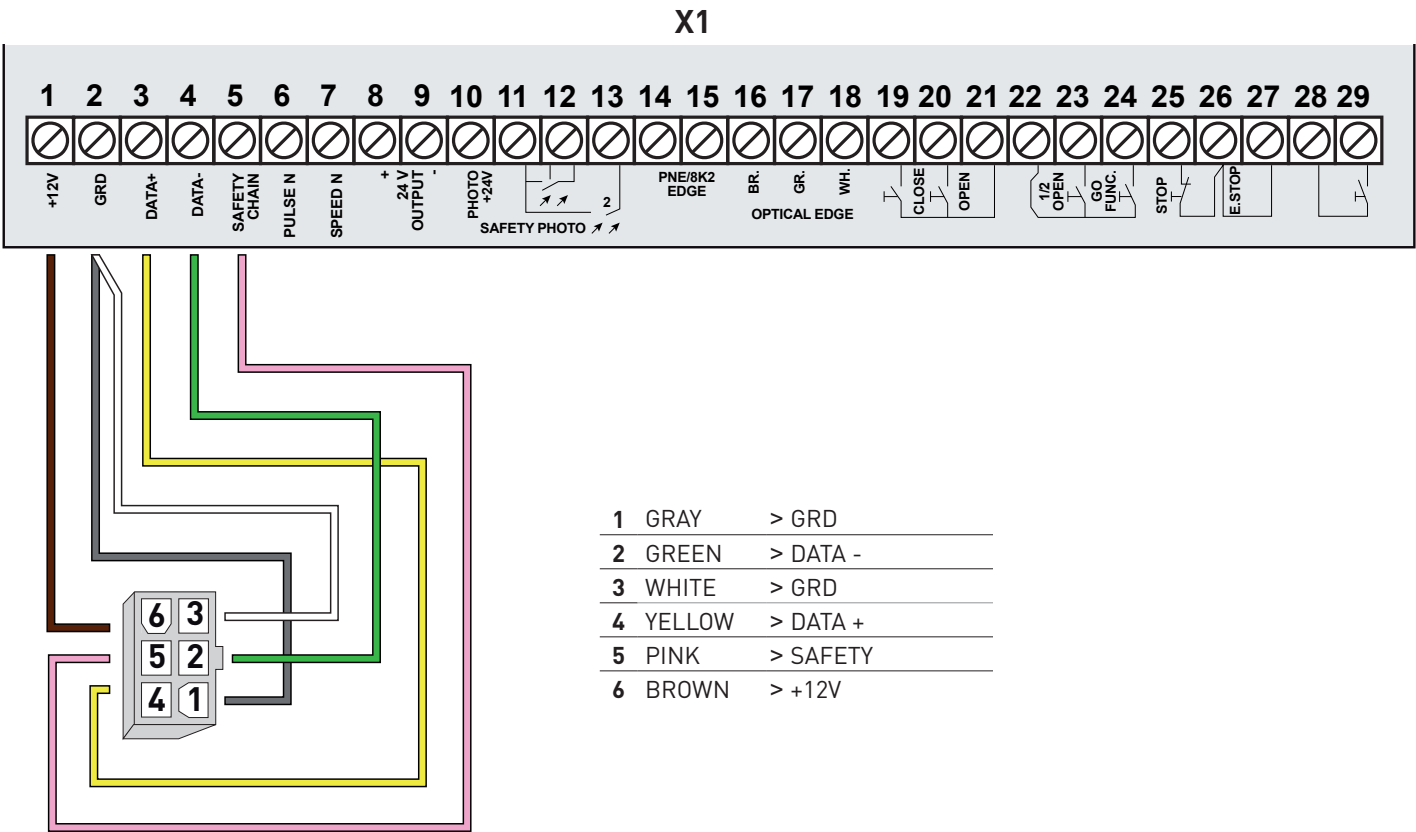
PAY ATTENTION AT THE CONNECTIONS! No line voltage (230V~ or other external devices) can be connected to the buttons otherwise the unit is damaged.

4 ELECTRONIC LIMITS WITH ENCODER

The encoder limits switch wires are connected according to the diagram of fig. 3.

ATTENTION: if you connect a control unit pre-set for mechanical limits to a motor with encoder limits, the motor does not perform correctly. In particular, the motor will not find the limit positions and this could put at risk the safety of people and/or things.

FIG. 3



5 CONTROL UNIT SET-UP

The set-up must be performed with the motor off. Follow carefully the steps as described in the procedures, DO NOT activate any safety, hand controls or radio controls unless specifically requested by the procedure.

Basically the set-up of the control and the right coupling control/motor must be performed by the installer.

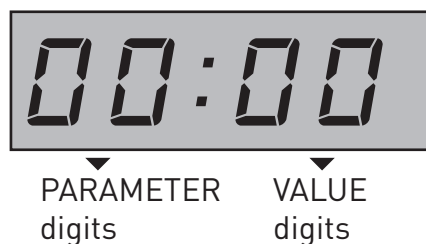
To enter the control unit programming mode place the DIP1 of the switch in ON.

During set-up the control unit will work only in dead man mode.

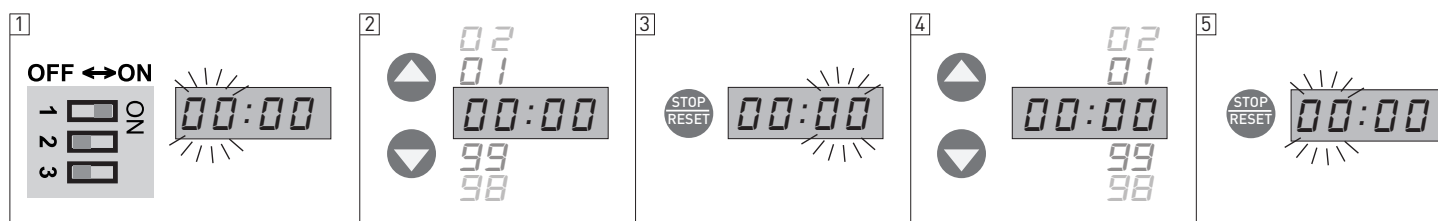
To return to the normal operating mode, place the DIP1 of the switch in OFF.

5.1 PARAMETER EXPLANATION

- STOP button  : to switch from PARAMETER field to VALUE field.
- OPEN  and CLOSE  buttons: to increase or decrease the size of the fields PARAMETER and VALUE



Note: shaded values correspond to the factory settings



STOP button: to switch from PARAMETER field to VALUE field.

OPEN / CLOSE buttons: to increase or decrease the size of the fields PARAMETER and VALUE

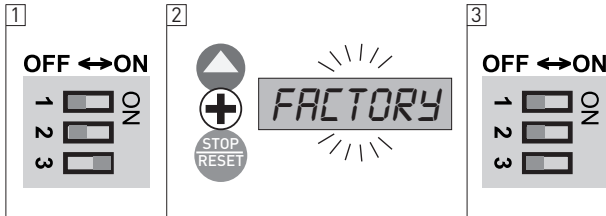
1. Put DIP switch 1 in ON position, PARAMETER digits start blinking
2. Select by OPEN / CLOSE buttons the number desired
3. Confirm by STOP button the PARAMETER selected. The VALUE digits start blinking
4. Select by OPEN / CLOSE buttons the number desired
5. To confirm the VALUE selected and return to PARAMETER field press STOP button
6. To leave the set-up mode, place the DIP1 in OFF.

Note: Some parameters require a further selection after pressing the STOP button as confirmation of the desired value. For example, to operate the door during the limit switch set-up the display shows **RUN**.

5.2 RESET PROCEDURE

The reset procedure allows to erase the settled data of the control unit memory and to return to the default programming.

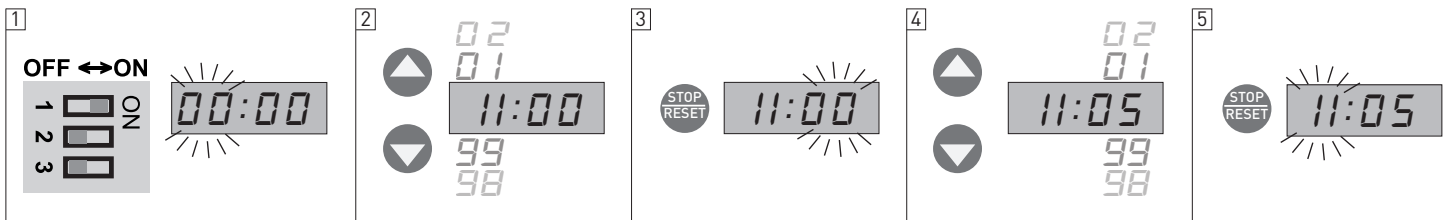
1. Put DIP 3 to ON position
2. Press simultaneously the STOP / OPEN buttons for 3 seconds at least
Display will shows **FACTORY** blinking and the control unit software version number.
3. Put DIP 3 to OFF position



6 SELECTION OF LIMIT SWITCH TYPES

SELECTION OF LIMIT SWITCH CODES

11:00	Not in use
11:01	Not in use
11:02	Not in use
11:03	Not in use
11:04	Not in use
11:05	Encoder limits - clockwise rotation
11:06	Encoder limits - anticlockwise rotation



1. Put DIP 1 in ON position, PARAMETER digits start blinking
2. Select by OPEN / CLOSE buttons the number 11
3. Confirm by STOP button the PARAMETER selected. The VALUE digits start blinking
4. Depending on the engine Gaposa used, select by OPEN / CLOSE buttons the VALUE:
 - VALUE 05: RAPIDO 30180 / 40130 / 6090 / 7045 / 60130
SZX 50 / 70 / 90 / 100 / 140 / 180
 - VALUE 06: SIDONE 65011 / 55011 / 40011 / 25012
RAPIDO 9090 / 12045
5. To confirm the VALUE selected and return to PARAMETER digits press STOP button
6. To leave the set-up mode, place the DIP1 in OFF.

WARNING: after selecting the type of digital limit switch with encoder it is necessary to cut off the power supply (by disconnecting the plug or by turning OFF the main switch) and then to connect it once again in order to allow the communication between the encoder and the control unit.

IMPORTANT NOTES

Please follow the installation requirements of the Gaposa motors. For example, if an encoded motor is installed in a way which the encoder direction is reversed, it will not run correctly and may put at risk things and/or people.

Gaposa disclaims any responsibility from the consequences of an installation not accomplished according to this policy.

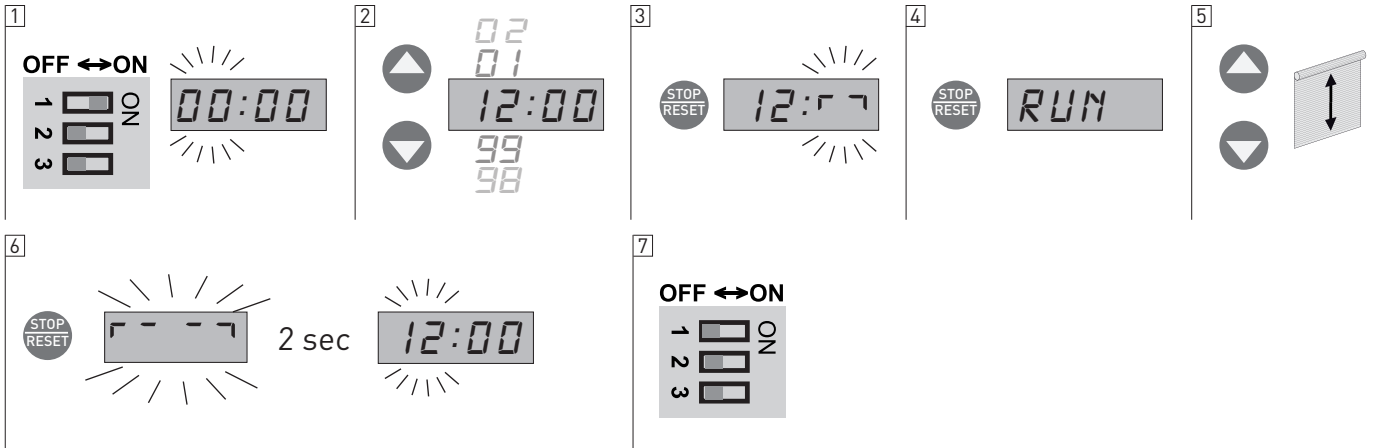
After the selection of the encoder limit switch, it is necessary to proceed with the limit switches adjustment (see section 7).

7 LIMIT SWITCH ADJUSTMENTS

Note:

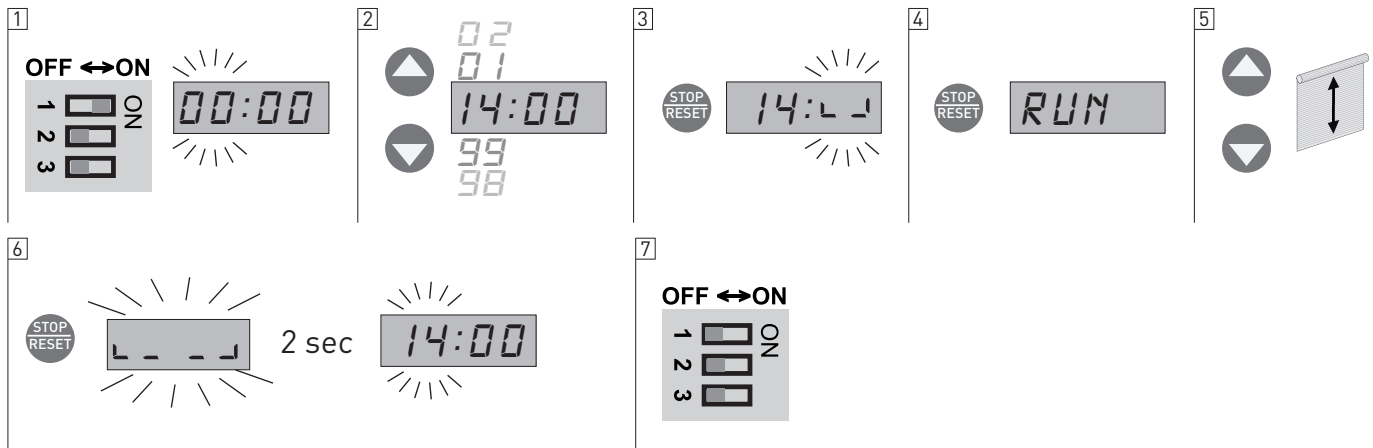
- the INTERMEDIATE POSITION function cannot be activated during programming (parameter 16) - Paragraph 12;
- the additional photocell on the door frame cannot be active during programming (parameter 31);
- When you change the limit switch positions the force control value (parameter 41) and operating time (parameter 51) will be reset to the factory default settings.

7.1 UP LIMIT SWITCH ADJUSTMENT



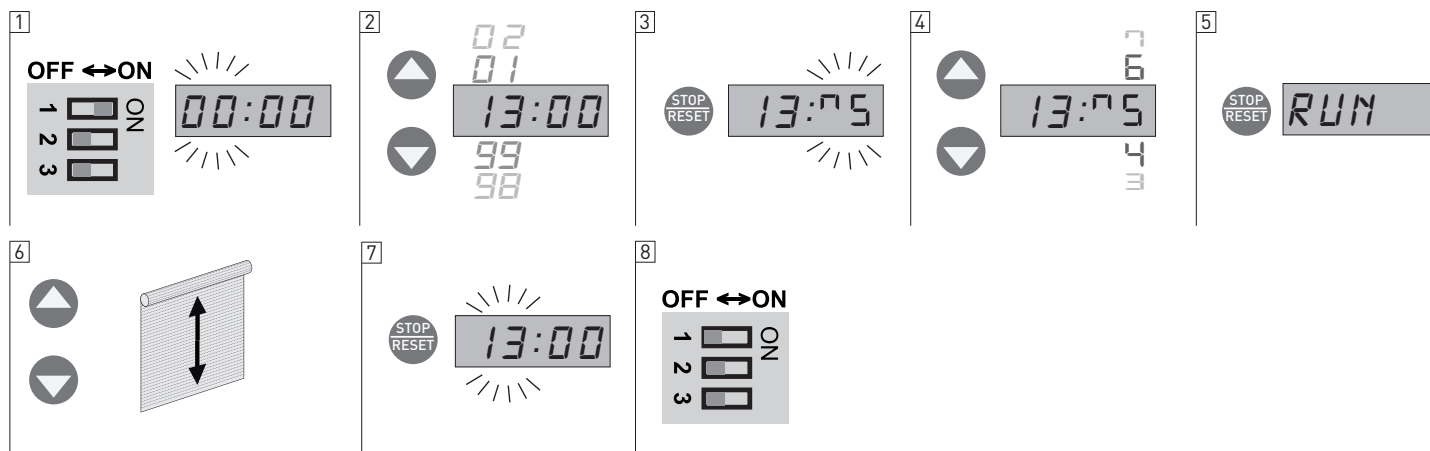
1. Put DIP 1 in ON position, PARAMETER digits start blinking
2. Using the buttons OPEN and CLOSE select the parameter 12
3. Access the field VALUE pressing STOP . The field VALUE shows the flashing symbol
4. Press button STOP once again and the unit, showing the message RUN , is ready to move the door
5. Use buttons OPEN and CLOSE to reach the exact UP limit position.
6. Press the STOP button to confirm the position. The display will show the symbol for 2 seconds and then the PARAMETER field will start flashing again (showing the number 12).
7. To leave the set-up mode, place the DIP1 in OFF.

7.2 DOWN LIMIT SWITCH ADJUSTMENT



1. With the DIP1 of the S4 switch in ON and the PARAMETER field still blinking
2. Select by OPEN / CLOSE buttons the parameter 14
3. Access the field VALUE pressing STOP . The field VALUE shows the flashing symbol
4. Press button STOP once again and the unit, showing the message RUN , is ready to move the door
5. Use buttons OPEN and CLOSE to reach the exact DOWN limit position.
6. Press the STOP button to confirm the position. The display will show the symbol for 2 seconds and then the PARAMETER field will start flashing again (showing the number 14).
7. To leave the set-up mode, place the DIP1 in OFF.

7.3 UP LIMIT SWITCH POSITION TUNING



1. Put DIP 1 in ON position, PARAMETER digits start blinking
2. Select parameter 13 using the buttons OPEN and CLOSE
3. Access the field VALUE pressing STOP. In the field VALUE symbol **~5** flashes
4. Using the buttons OPEN and CLOSE vary the value:

Value from 4 to 1: progressively decrease the UP position

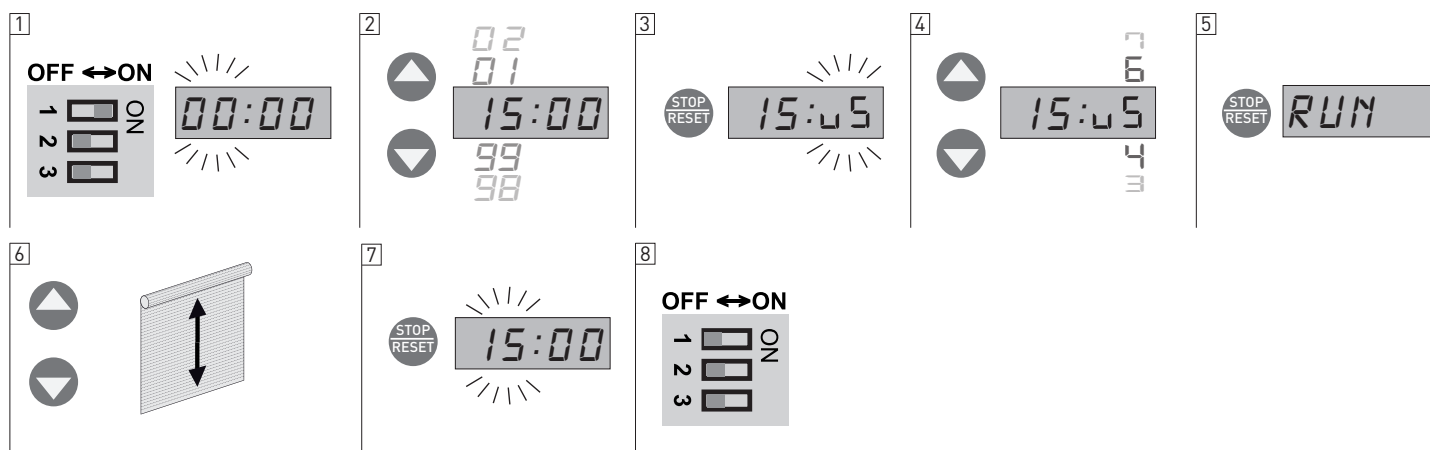
Value from 6 to 9: progressively increase the UP position.

The adjustment range is max $\pm 0.8\%$ of the travel of the door.

If the value is not to be changed you can return to the field PARAMETER pressing the STOP button

5. After modifying the VALUE press the STOP button to confirm: the display will show **RUN**
6. You can test the varied position of the door by moving it through the buttons OPEN and CLOSE.
7. Press the STOP button once again to confirm the tuning and return to the PARAMETER field.
8. To leave the set-up mode, place the DIP1 in OFF.

7.4 DOWN LIMIT SWITCH POSITION TUNING



1. Put DIP 1 in ON position, PARAMETER digits start blinking
2. Select parameter 15 using the buttons OPEN and CLOSE
3. Access the field VALUE pressing STOP . In the field VALUE symbol 05 flashes
4. Using the buttons OPEN and CLOSE vary the value:

value from 4 to 1: progressively decreases the DOWN position

value of 6 to 9: progressively increases the DOWN position.

The adjustment range is max $\pm 0.8\%$ of the travel of the door.

If the value is not to be changed you can return to the field PARAMETER pressing the STOP button

5. After modifying the VALUE press the STOP button to confirm: the display will show RUN
6. You can test the varied position of the door by moving it through the buttons OPEN and CLOSE .
7. Press the STOP button once again to confirm the tuning and return to the PARAMETER field.
8. To leave the set-up mode, place the DIP1 in OFF.

8 OPERATION MODE

The control unit is pre-set in dead-man control mode (PARAMETER 01, VALUE 01).

It is possible, however, to define different working modes by modifying the value of PARAMETER 01:

01:01 Hold-to-run OPEN
Hold-to-run CLOSE (Put a bridge in X1 terminal 14-15 when there is no safety device)

01:02 Impulsive OPEN
Hold-to-run CLOSE (Put a bridge in X1 terminal 14-15 when there is no safety device)

01:03 Impulsive OPEN; Impulsive CLOSE.

01:04 Not in use

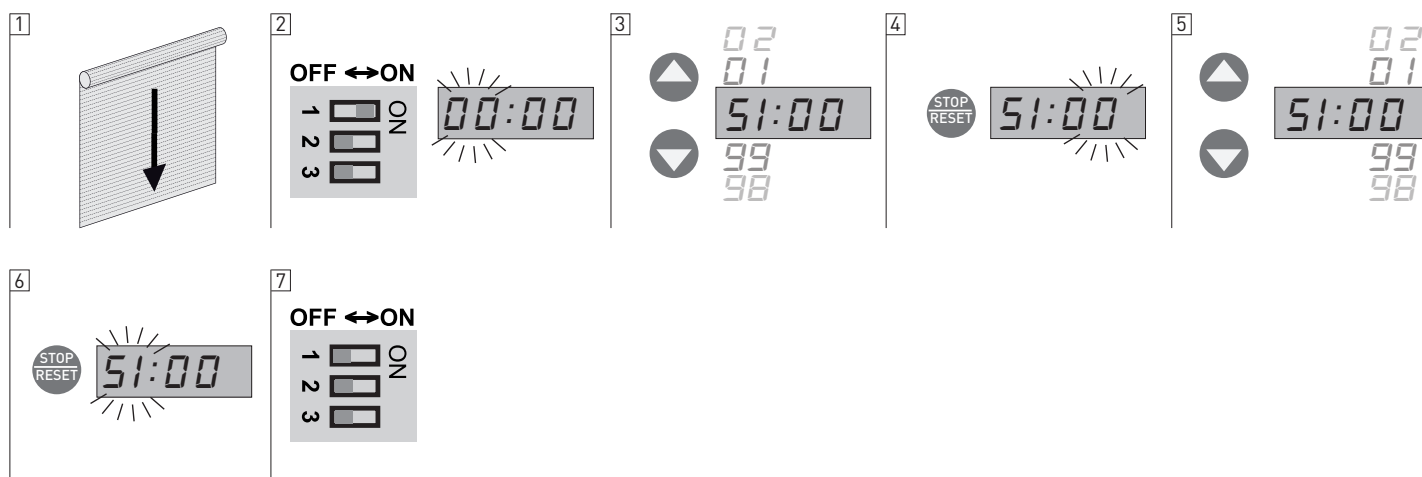
During the encoder limit switches set-up the control unit will only allow the dead-man working mode.

9 WORKING TIME SET-UP

PARAMETER 51 defines the working time of the door.

WARNING! The default parameter is the **51:02** that is to say a working time of 40 seconds

To turn off or modify the working time, follow this procedure:



1. Close the door and stop at the DOWN limit position.
2. Put DIP 1 in ON position, PARAMETER digits start blinking
3. Using the buttons OPEN and CLOSE select the parameter 51
4. Access the field VALUE pressing STOP
5. Using the buttons OPEN and CLOSE vary the value

value 00: Function inactive

value 01: Working time 20 seconds

value 02: Working time 40 seconds (default)

value 03: Activate the self learning function to determine the working time.

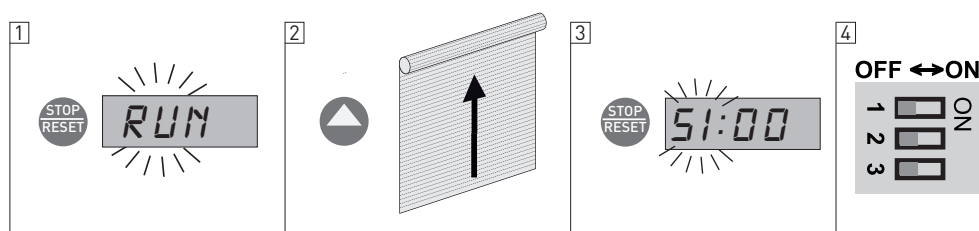
Caution: In order to use this function the limit switches must be already adjusted.

value 04: Working time 60 seconds

5a. Select the value 00 / 01 / 02 / 04 then press STOP to confirm

6a. Place the DIP1 again in OFF to be out of the set-up mode

*** If value 03 selected:**



6b. Press STOP to confirm. The control unit, showing **RUN** is ready to move the door

7b. Using the OPEN button move the door from the closed position to the open position without interruptions.

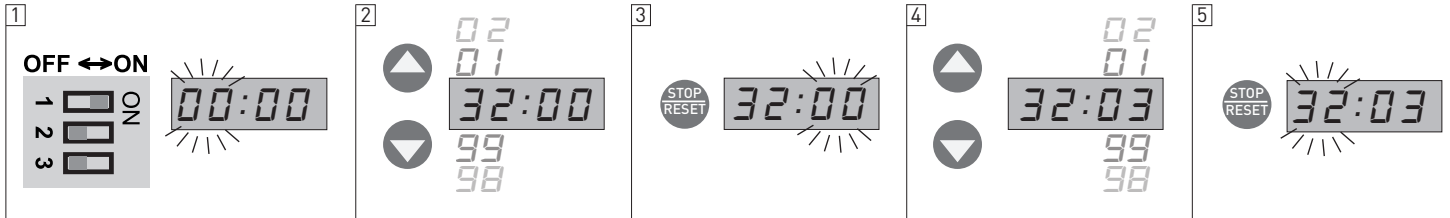
8b. Once the UP limit switch is reached, the door stops, **RUN** stops flashing and the display will automatically return to field PARAMETER.

9b. To leave the set-up mode, place the DIP1 in OFF

10 AUTOMATIC CLOSING

Parameter 32 allows the selection of the door automatic closing after a selectable period of time.

Important: parameter 32 is visible and accessible only if parameter 01 has been set in impulsive mode **01:03**



1. Put DIP 1 in ON position, PARAMETER digits start blinking
2. Using the buttons OPEN and CLOSE select the parameter 32
3. Access the field VALUE pressing STOP
4. Using the buttons OPEN and CLOSE vary the value
 - The value 00 in the field VALUE inhibits the automatic closing;
 - A value greater than 0, from 1 to 990, indicates the number of seconds to wait before the activation of the automatic closing:

NOTE: From 0 to 99 the change is made every second by using the buttons OPEN and CLOSE.

Over 99 the change is made every 10 seconds and the value will flash quickly: for example, the VALUE 18 corresponds to 180 seconds, the value 19 to 190 seconds ...

If you keep the OPEN button pressed you will get a fast increase of the value

5. Press STOP to confirm
6. To leave the set-up mode, place the DIP1 in OFF.

Warning: the interlock function prevents the automatic closing when activated. For details see section 11 (INTERLOCK)

11 INTERLOCK

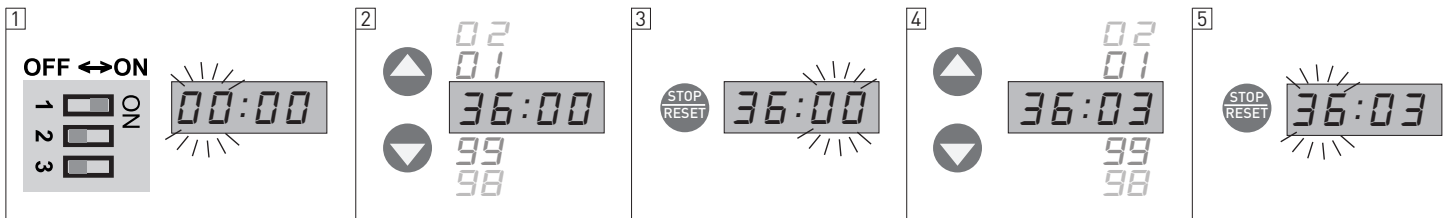
The Interlock function stops the automatic closing if activated, to prevent a closing of the door.

The countdown on the display shows the value of the pre-set waiting time. To activate the Interlock function, with the door in its UP limit position, hold the STOP button or the emergency stop button for more than 5 seconds.

To deactivate the interlock mode press CLOSE button. In case you want to disable the interlock function, choose the value 00 in parameter 36.

ATTENTION! Parameter 36 is visible and selectable only if the automatic closing has been selected in parameter 32.

To enable the Interlock function:



1. Put DIP 1 in ON position, PARAMETER digits start blinking
2. Using the buttons OPEN and CLOSE select the parameter 36
3. Access the field VALUE pressing STOP
4. Using the buttons OPEN and CLOSE vary the value
 - value 00: interlock function OFF;
 - value 01: interlock function ON;
5. Press STOP to confirm
6. To leave the set-up mode, place the DIP1 in OFF.

12 INTERMEDIATE POSITION

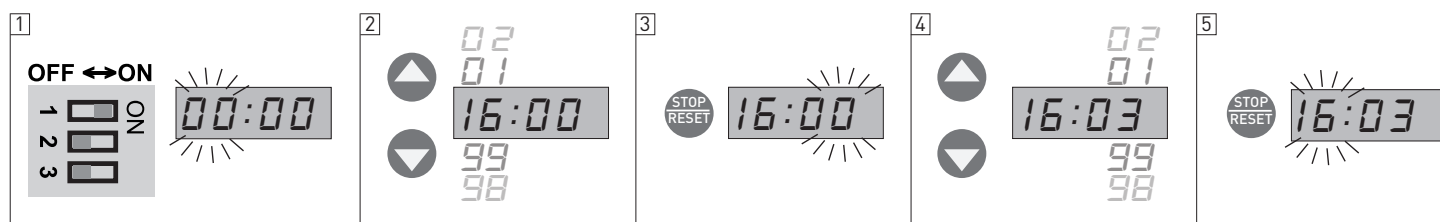
The intermediate position can occur through the use of a selector or of an additional button. In this case the PARAMETER 16 must be selected at VALUE 00.

If you are using a selector, this one must be connected to terminals 22 and 23 of X1.

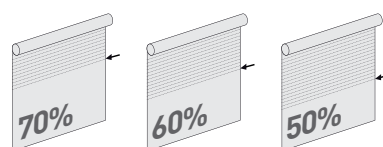
If you open the contact of the selector, the use of the intermediate position is inhibited.

If you close the contact of the selector, pressing the UP button, the door stops at the intermediate position.

The intermediate position can be adjusted by setting the PARAMETER 16 at values from 02 to 07 with a progressive change of the intermediate position from 50% to 75% of the travel.



1. Put DIP 1 in ON position, PARAMETER digits start blinking
2. Using the buttons OPEN and CLOSE select the parameter 16
3. Access the field VALUE pressing STOP
4. Using the buttons OPEN and CLOSE vary the value
 - VALUE 02: Intermediate position at 50% of the travel
 - VALUE 03: Intermediate position at 55% of the travel
 - VALUE 04: Intermediate position at 60% of the travel
 - VALUE 05: Intermediate position at 65% of the travel
 - VALUE 06: Intermediate position at 70% of the travel
 - VALUE 07: Intermediate position at 75% of the travel
5. Press STOP to confirm



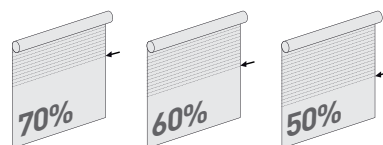
If you are using an additional button, this one must be connected to terminals 22 and 23 of X1.

In this case the UP button allows the opening of the door up to the UP limit position.

While, to move the door to the intermediate position you have to press the additional button.

The intermediate position can be determined by setting the PARAMETER 16 with values from 08 to 13 with a progressive change of the intermediate position from 50% to 75% of the travel:

- VALUE 08: Intermediate position at 50% of the travel
- VALUE 09: Intermediate position at 55% of the travel
- VALUE 10: Intermediate position at 60% of the travel
- VALUE 11: Intermediate position at 65% of the travel
- VALUE 12: Intermediate position at 70% of the travel
- VALUE 13: Intermediate position at 75% of the travel



12.3 AUTOMATIC CLOSING FROM INTERMEDIATE POSITION

You can set the automatic closing even from the intermediate position setting the PARAMETER 17.

1. Activate the automatic closing function (paragraph 10)
2. Set the PARAMETER 17:
 - VALUE 00: Automatic closing (from intermediate position) OFF
 - VALUE 01: Automatic closing (from intermediate position) ON

13 SAFETY DEVICES

13.1 SAFETY BEAMS

Refer to the safety beams instructions for the DC supply.

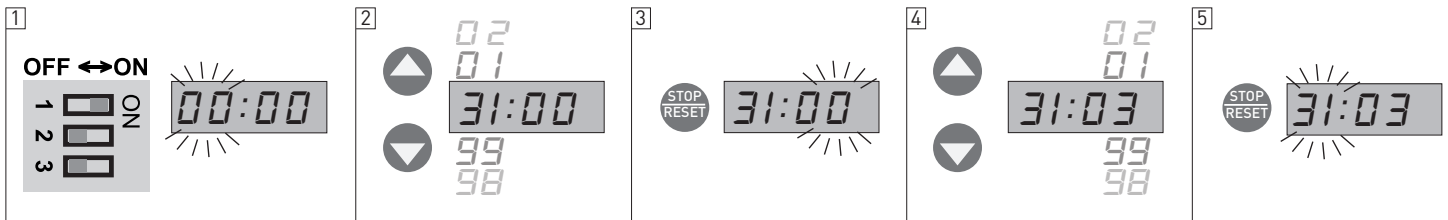
Safety beams wiring:

- terminal 10 [PHOTO +24V] power supply / terminal 9 [-] GRD.
- terminals [11-12] NC photo 1 signal
- terminals [11-13] NC photo 2 signal

ATTENTION! Both the transmitter and the receiver of the safety beams must be connected to the same terminals.

Through parameter 31 you can conform the control unit to the type of connection that you are going to select, in order to activate the corresponding test functions.

This test allows the control unit to constantly check of short circuits or malfunctions that could compromise the safety of the device. The test thus allows to ensure the safety even in case of single failure as required by the standards EN13241-1 and EN-12453.



1. Put DIP 1 in ON position, PARAMETER digits start blinking
2. Using the buttons OPEN and CLOSE select the parameter 31
3. Access the field VALUE pressing STOP
4. Using the buttons OPEN and CLOSE vary the value:
 - VALUE 00: No safety beams connected
 - VALUE 01: Connect using safety beams 1
 - VALUE 02: Connect using safety beams 2
 - VALUE 03: Connect using safety beams 1 and 2
5. Press STOP to confirm
6. To leave the set-up mode, place the DIP1 in OFF.

13.2 SAFETY EDGE

Safety edge connection: in case of resistive safety edges type 8k2 ohm or pneumatic safety edges, connect the wires to the terminals 14 and 15 of X1;

In case of optoelectronic safety edge, connect the wires to the terminals 16, 17 and 18 of X1 (respecting the color order).

Warning: if you choose the optical safety edge (VALUE 03) the terminals 14 and 15 DO NOT have to be connected by a jumper.

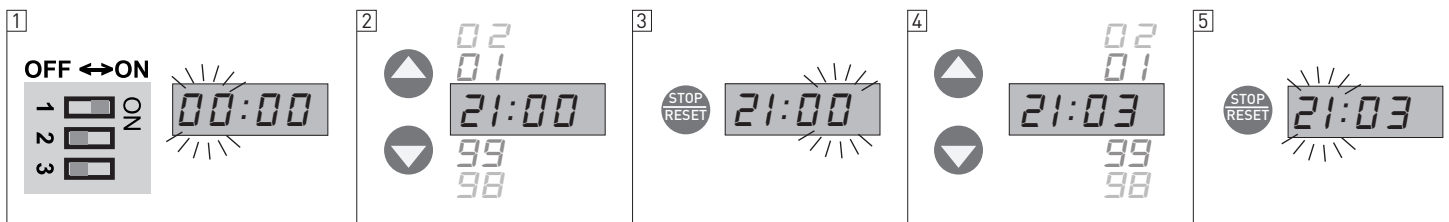
Warning: if you DO NOT want to use a safety edge, select the VALUE 01 and connect the terminals 14 and 15 with a jumper. The terminals 16, 17 and 18 of X1 must not be connected.

Warning: the safety edge must be connected before the selection of PARAMETER 21, but do not activated them.

If this happens, the control unit shows an error signal on the display the code **EFF**

The same happens if you choose a parameter that does not match with the connected terminals.

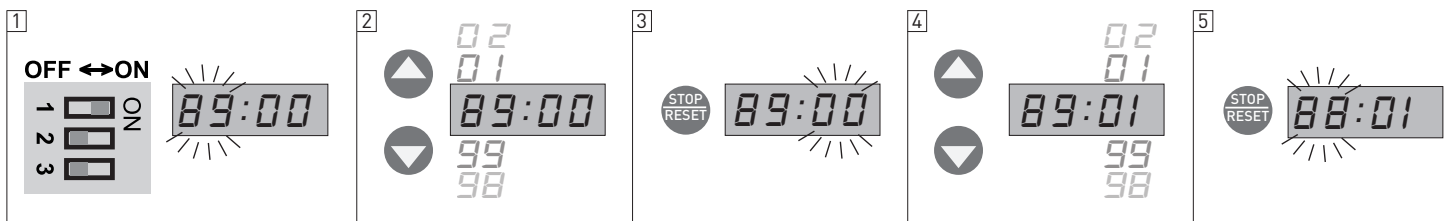
Through PARAMETER 21 you can select the type of safety edge.



1. Put DIP 1 in ON position, PARAMETER digits start blinking
2. Using the buttons OPEN and CLOSE select the parameter 21
3. Access the field VALUE pressing STOP
4. Using the buttons OPEN and CLOSE vary the value
 - VALUE 01: PNE / DW pneumatic
 - VALUE 02: Safety edge with resistive contact 8k2 ohm
 - VALUE 03: Optoelectronic edge
 - VALUE 04: Special LP / DW pneumatic
5. Press STOP to confirm
6. To leave the set-up mode, place the DIP1 in OFF.

13.3 AUX 3 (solid state relays)

Through PARAMETER 89 it is possible to determine the behavior of AUX relé (solid state).



- Put DIP 1 in ON position, PARAMETER digits start blinking
2. Using the buttons OPEN and CLOSE select the parameter 89
 3. Access the field VALUE pressing STOP
 4. Select the preferred value by pressing buttons S2 and S3:
 - VALUE 00: deactive relé
 - VALUE 01: active relé
 5. Press STOP to confirm
 6. To leave the set-up mode, place the DIP1 in OFF

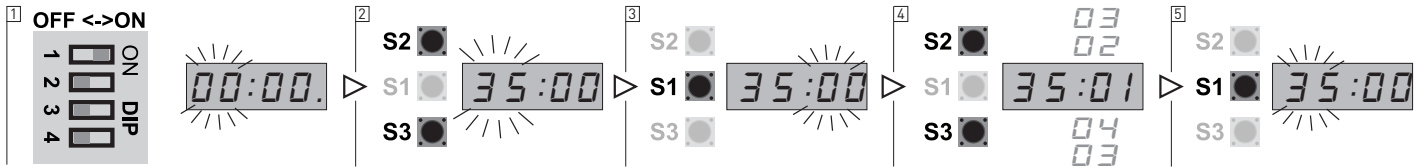
14 ADDITIONAL ACCESSORIES

14.1 ADDITIONAL RADIO RECEIVER SLOT QCMR500 (OPTIONAL)

The control can be radio operated thanks to the QCTE transmitter that allows the storage of 1997 radio codes. The QCTE transmitter must be matched to the QCMR500 radio receiver slot, already connected to slot X7 (see page 6).

Please look at the instructions attached to the QCMR500 slot in order to connect it to the control unit and to match it to the transmitter.

Once the QCMR500 slot is inserted, you can set up its operation mode through **PARAMETER 35** (parameter 35 is visible **ONLY** if the photocell is active through parameter 31):



1. Put DIP 1 (S4) in ON position, **PARAMETER** digits start blinking
2. Using the buttons **OPEN** (S2) and **CLOSE** (S3) select the parameter 35
3. Access the field **VALUE** pressing **STOP** (S1)
4. Select the preferred value by pressing buttons S2 and S3:

- **VALUE 00. GENERAL MODE OF OPERATION**

the signal of the transmitter always commands the opening, except when the door is already completely opened. In this case it commands the closure.

- **VALUE 01. GENERAL MODE OF OPERATION + STOP**

the signal stops the movement of the door **ONLY** while opening

- **VALUE 02: MODE OF OPERATION "ONLY OPENING"**

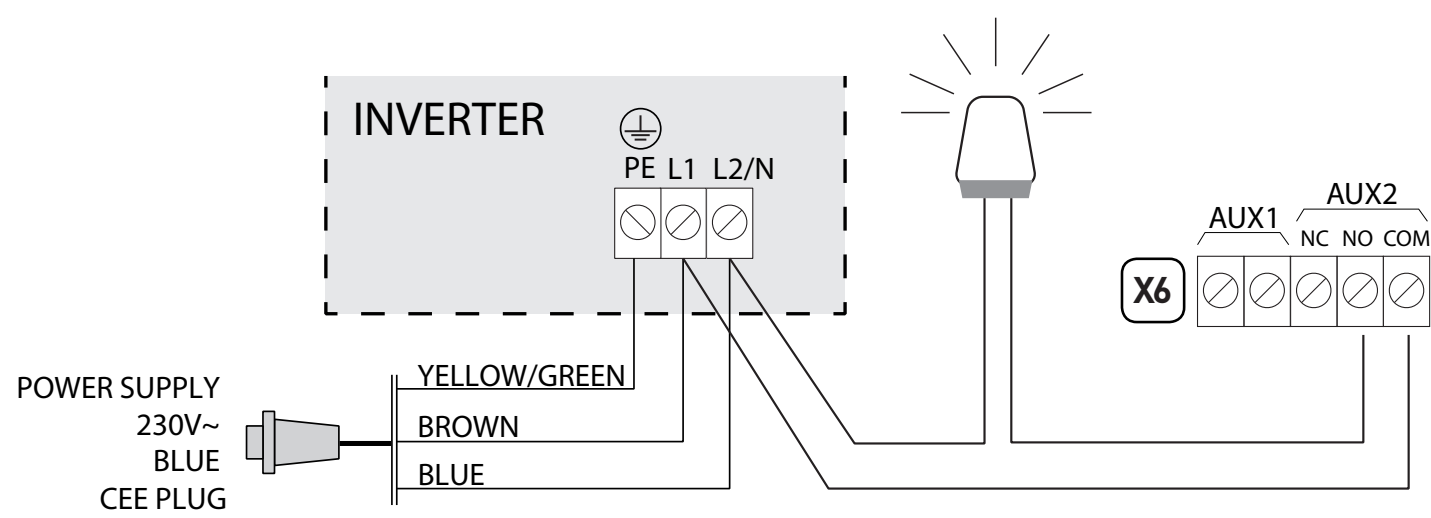
the signal of the transmitter only activates the opening of the door. If the door is closing, the signal changes the movement till the UP limit switch position has been reached.

NOTE: if the AUTOMATIC CLOSURE has been programmed, during the pause the signal of the transmitter will extend the pause by resetting the timer of the automatic closure.

5. Press **STOP** (S1) to confirm
6. To leave the set-up mode, place the DIP1 in OFF

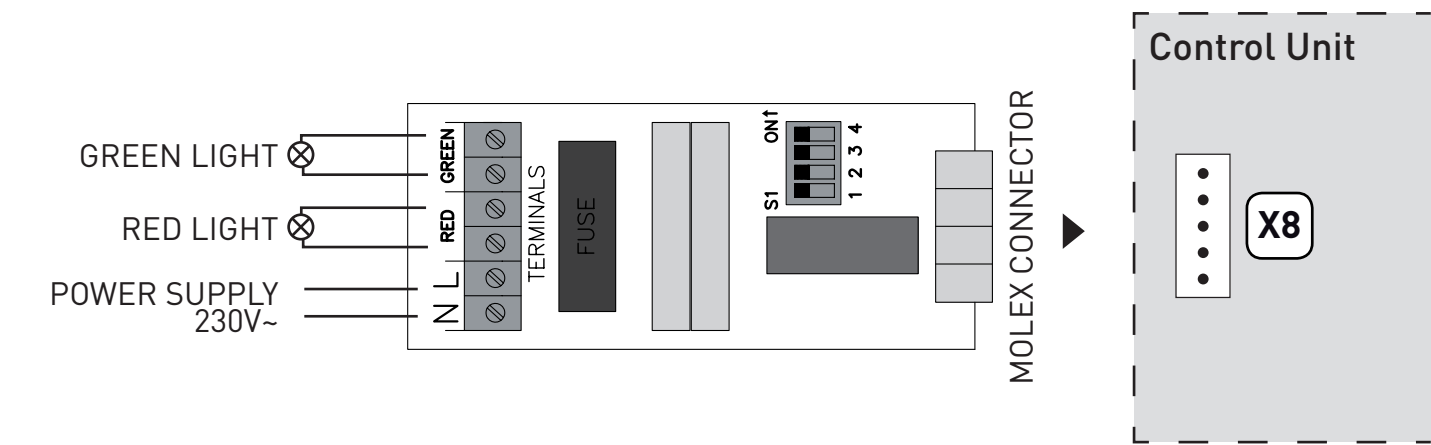
NOTE: the entrance "GO" (terminals 22-24 on X1) follows the same logical of functioning selected for radio receiver (parameter 35).

14.2 FLASHING LIGHT CONNECTION (230Vac with autolamping) / COURTESY LAMP



14.3 ADDITIONAL PLUG-IN BOARD TO CONTROL FLASHING LIGHTS, COURTESY LAMPS AND TRAFFIC LIGHTERS - QCLSM

QCLSM Plug-in board to be used in order to control flashing lights, courtesy lamps or traffic lighters during the device operation (door, shutter, etc.) controlled by the panel where it is installed. The possible operation options are described in the QCLSM instruction manual.



15 INVERTER SETTINGS

15.1.1 MOTOR SETTINGS

100: 1,1	Rated motor power [kW]
101: 230	Rated voltage [V]
102: 5,9	Rated current [A]
103:1400	Rated current [A]
104: 50	Rated frequency [Hz]
105: 0	Autotune Note! Future option
109: 100	Max. frequency [Hz] Note: If frequency max is adjusted below parameter 110, 111, 120 or 121 value these values is automatic reduced also.

15.1.2 SETTINGS OPENING DIRECTION

110: 50	Frequency HIGH speed opening [Hz]
111: 10	Frequency LOW speed opening [Hz]
112: 1,0	Acceleration time opening [Sec.]
113: 1,0	Deceleration time opening, high speed to low speed [Sec.]
114: 0,3	Deceleration time to stop [Sec.]
115: 30	Low speed set point before open limit [%] of full distance.

15.1.3 SETTINGS CLOSING DIRECTION

120: 30	Frequency HIGH speed closing [Hz]
121: 10	Frequency LOW speed closing [Hz]
122: 1,0	Acceleration time closing [Sec.]
123: 2,0	Deceleration time closing, high speed to low speed [Sec.]
124: 0,3	Deceleration time to stop [Sec.]
125: 30	Low speed set point before close limit [%] of full distance.

Note! All acceleration/deceleration times, are seconds from 0 Hz → max Hz / max Hz → 0 Hz.
(Max Hz = parameter 109)

DISPLAY IN RUN MODE

The display will in run mode show status of limits, some inputs or error codes if they occurs.
When power up the software version is showed shortly.

DISPLAY	DESCRIPTION
 STANDBY	- Nothing active. (4 chairs symbol) - Door is stopped between limits and no errors are found.
 GATE OPEN	Open limit active
 GATE CLOSED	Close limit active
 HALF OPEN	1/2 open limit active
 STOPPED	Stop attivo
 OPEN	OPEN push-button active
 CLOSE	CLOSE push-button active
 GO FUNCTION	GO function active (Note that the door only can be closed by GO function, when photo is installed)
 PHOTO 1 ACTIVATED	Photo 1 active Photo 2 is external photo cells mounted in the screw terminals X12.
 PHOTO 2 ACTIVATED	Photo 2 active Photo 2 is external photo cells mounted in the screw terminals X3.
 SAFETY EDGE ACTIVATED	Safety Edge active
 SETUP ERROR	Safety list not mounted correct / wrong selection in parameter 21.
 OPENING	Door running up
 CLOSING	Door running down

DISPLAY	DESCRIPTION
Service required	Error code. Door is running without command Service needed. Fatal error. Move the door manual to middle position without power. Change from normal mode to programming mode on DIP switch no. 1. This will clear the SER error. If the door is running again in 1 sec. without command when power is on then the PCB is defect.
E:01 Edge fail	Error code. Edge monitoring Error code Monitoring failure of safety edge if this function is activated. Check or adjust safety edge list. See chapter 1.5.
E:03 Run time	Error code. Run time Error code. Door is stopped on run time control. See chapter 1.10.
E:04 Service counter expired	Error code. Service Service counter decremented to 0 Reset for new countdown
E:05 Photo circuit fail	Error code. Photo Failure in photo circuit. (Test cycle after last stop failed, Press stop to start new test)
E:06 Safety Edge fail	Error code. Safety Edge Failure in edge circuit. (Test cycle after last stop failed, Press stop to start new test)
E:09 No Position change	Error code. no change of encoder position, when running. Door started, but the position is not changing. Door is stopped after delay time and E:09 failure is shown about 1 sec. Possible errors: The door is blocked, disengaged, cable connection error or the encoder magnet is not fixed on the shaft. Reset of E09: both limits shall be founded again by hold-to-run steps. (If it is not possible to find both limits, the limits must be relearned) If necessary, adjust in parameter 81 (delay time) (Parameter 81:03 = autoreset)
E:20 EEPROM fail	Error code. EEPROM Fail Possible error: Limits has been changed, after the force control has been learned. Reset of E20: Try deactivating force control in parameter 41 (41:00) and after this make a new power-up.
E:21 EEPROM PWR-UP fail	Error code. EEPROM Fail EEPROM failure of power-up. Try factory clear or change processor.
Error 12V or 24V	Error on 24V and/or 12V voltage circuit. 24/12V is shorted or overloaded.

PARAMETER LIST:

01:01	Operation settings
02:00	Special setting regarding failure on safety list or photo
11:00	Selection of limits (Mechanical/encoder)
12: 7 7	Electronic limit settings, OPEN
13: 5	Electronic limit settings, OPEN fine tuning
14: L J	Electronic limit settings, CLOSE
15: 5	Electronic limit, CLOSE fine tuning
16:00	1/2 Open settings
17:00	Auto close 1/2 open
21:01	Selection of safety list
22:00	Electronic after run
23:00	Extra safety list
29:00	Wire tightening function
31:00	Photo cells
32:00	Auto clos
33:00	Car wash function
34:00	"Forced" closing
35:00	G0 function (Step)
36:01	Interlock function ON/OFF
41:00	Force control – Section
42:60	Force control MANUEL OPEN settings
43:50	Force control MANUEL CLOSE settings
44:00	Force control automatic settings (41:03)
51:00	Run time
52:01	Reverse time – safety list
53:01	Reverse time – photo cells
58:00	Service counter
59:00	Service counter reaction
81:03	Encoder position failure
82:01	Encoder test function
84:00	Special open function
88:00	Relay K3 settings (option) Reset to factory settings



Gaposa srl - via Ete, 90 - 63900 Fermo - Italy
T. +39.0734.220701 - F. +39.0734.226389 - info@gaposa.com
www.gaposa.com

QC600_QC600S_EN_0319